

QUESTION PAPER CODE: VW43

Name of the Candidate:

Reg. No.

NOVEMBER 2025 ARREAR EXAMINATIONS

B.E. ELECTRICAL AND ELECTRONICS ENGINEERING

22EE410 PROBABILITY DISTRIBUTION AND RANDOM PROCESS

Duration: 3Hours

Maximum: 100 Marks

SPECIAL INSTRUCTIONS: Students are permitted to use Statistical tables.

Answer ALL Questions

PART – A

(5 x 6 = 30)

CO

Marks

- A1.** Let X be the rated capacity of a freezer of this brand sold at a certain store. Suppose that X has probability mass function as:

CO1

(6)

x	16	18	20
p(x)	0.2	0.5	0.3

(a). Compute E(X) and V(X).

(b). If $h(X) = 70X - 650$, then compute the expected value and variance of h(X).

- A2.** The joint distribution of X and Y is given in the following table:

CO2

(6)

Y \ X	0	1	2	3
0	0.08	0.07	0.04	0
1	0.06	0.15	0.05	0.04
2	0.05	0.04	0.10	0.06
3	0	0.03	0.04	0.07
4	0	0.01	0.05	0.06

Compute (a). $P(X = 1 \text{ and } Y = 1)$ (b). Marginal probability mass function of X and Y.

- A3.** A random sample of 150 recent donations at a certain blood bank reveals that 82 were type A blood. Test the hypotheses that the actual percentage of type A donations differs from 40%, the percentage of the population having type A blood using a significance level of 0.01.

CO3

(6)

- A4.** Calculate a 99% confidence interval for the difference between population mean total cholesterol level for vegans and population mean total cholesterol level for omnivores.

CO4

(6)

Diet	Sample Size	Sample Mean	Sample SD
Vegan	88	5.10	1.07
Omnivore	93	5.55	1.10

- A5.** A stationary process has an Auto correlation function of $\frac{25\tau^2 + 36}{6.25\tau^2 + 4}$. Compute mean and variance of the process.

CO5

(6)

PART – B

(70 Marks)

CO

Marks

- B1.** a. Suppose the force acting on a column that helps to support a building is a normally distributed random variable X with mean value 15.0 kips and standard deviation 1.25 kips. Compute the following probabilities.

CO1

(5)

(i). $P(X \leq 17.5)$ (ii). $P(|X - 15| \leq 3)$.

- b. Let X denote the distance (m) that an animal moves from its birth site to the first territorial vacancy it encounters. Suppose that for banner-tailed kangaroo rats, X has an exponential distribution with parameter $\lambda = 0.01386$

CO1

(5)

(i). Calculate the probability that the distance is at most 100 m.

(ii). Calculate the probability that the distance is between 100 and 200 m.

(OR)

- B2.** a. Compute the following binomial probabilities: CO1 (5)
(i). $b(3; 8, 0.35)$ (ii). $P(2 \leq X \leq 4)$ when $n = 7$ and $p = 0.6$.
- b. Suppose that the number of drivers travel between a particular origin and destination during a designated time period has a Poisson distribution with parameter $\mu = 20$. Compute the probability that the number of drivers will CO1 (5)
(a). Be at most 10. (b). Exceed 20. (c). Be between 10 and 20 inclusive.

- B3.** The joint probability density function of the random variables X and Y is given CO2 (15)
by $f(x, y) = \begin{cases} 2 - x - y & , 0 \leq x \leq 1, 0 \leq y \leq 1 \\ 0 & , \text{otherwise} \end{cases}$
(i). Compute $E(X)$ and $E(Y)$. (ii). Compute the covariance of X and Y
(iii). Compute the correlation coefficient of X and Y.

(OR)

- B4.** a. Suppose the actual air pressure in each tire is a random variable X for the right tire and Y for the left tire, with joint pdf CO2 (8)
 $f(x, y) = \begin{cases} k(x^2 + y^2) & , 20 \leq x \leq 30, 20 \leq y \leq 30 \\ 0 & , \text{Otherwise} \end{cases}$
(i). Compute the value of k (ii). Show that X and Y are independent.
- b. An instructor has given a short quiz consisting of two parts. For a randomly selected student, let X be the number of points earned on the first part and Y be the number of points earned on the second part. Suppose that the joint probability mass function of X and Y is given in the accompanying table. CO2 (7)

P(x,y)		y			
		0	5	10	15
x	0	0.02	0.06	0.02	0.10
	5	0.04	0.15	0.20	0.10
	10	0.01	0.15	0.14	0.01

Compute the covariance of X and Y.

- B5.** a. To obtain information on the corrosion-resistance proper ties of a certain type of steel conduit, 45 specimens are buried in soil for a 2-year period. The maximum penetration (in mils) for each specimen is then measured, yielding a sample average penetration of mean = 52.7 and a sample standard deviation of $s = 4.8$. The conduits were manufactured with the specification that true average penetration be at most 50 mils. They will be used unless it can be demonstrated conclusively that the specification has not been met. Carry out a test of the relevant hypotheses using a significance level of 0.05. CO3 (5)
- b. Reconsider the accompanying sample data on expense ratio (%) for large-cap growth mutual funds first introduced CO3 (6)
0.52 1.06 1.26 2.17 1.55 0.99 1.10 1.07 1.81 2.05 0.91 0.79
1.39 0.62 1.52 1.02 1.10 1.78 1.01 1.15
normal probability plot shows a reasonably linear pattern. Is there compelling evidence for concluding that the population mean expense ratio exceeds 1%? Carry out a test of the relevant hypotheses using a significance level of .01

(OR)

- B6.** a. An article reported on a pilot study in which each of 58 individuals in a sample was asked to estimate the calorie content of a 12-oz can of beer known to contain 153 calories. The resulting sample mean estimated calorie level was 191 and the sample standard deviation was 89. whether this data suggest that the true average estimated calorie content in the population sampled exceeds the actual content? Test the appropriate hypotheses at significance level .001. CO3 (5)

- b. The melting point of each of 16 samples of certain brand of hydrogenated vegetable oil was determined, resulting in $\bar{x} = 94.32$. Assume that the distribution of the melting point is normal with $\sigma = 1.20$, Test $H_0: \mu = 95$ versus $H_a: \mu \neq 95$ using a level 0.01. CO3 (6)

- B7.** a. Two different types of alloy, A and B, have been used to manufacture experimental specimens. The ultimate strength (ksi) of each specimen was determined, and the results are summarized in the accompanying frequency distribution. CO4 (7)

	A	B
26 - < 30	6	4
30 - < 34	12	9
34 - < 38	15	19
38 - < 42	7	10

$$m = 40 \quad n = 42$$

Calculate a 95% CI for the difference between the true proportions of all specimens of alloys A and B that have an ultimate strength of at least 34 ksi.

- b. The following table gives the classification of 100 workers according to sex and nature of work. Test whether the nature of the work is independent of the sex of the worker using χ^2 - value. CO4 (6)

	Stable	Unstable
Male	40	20
Female	10	30

(OR)

- B8.** The article reported the data on daily calorie intake both for a sample of teens who said they did not typically eat fast food and another sample of teens who said they did usually eat fast food. CO4 (13)

Eat Fast Food	Sample Size	Sample Mean	Sample SD
No	663	2258	1519
Yes	413	2637	1138

Whether this data provide strong evidence for concluding that average calorie intake for teens who typically eat fast food exceeds by more than 200 calories per day the true average intake for those who don't typically eat fast food? Investigate by carrying out a test of hypotheses at a significance level of .05.

- B9.** If $X(t) = \sin(\omega t + Y)$ where Y is uniformly distributed in $(0, 2\pi)$. Show that $X(t)$ is wide sense stationary. CO5 (11)

(OR)

- B10.** a. Consider the random process $X(t) = \cos(t + \theta)$ where θ is uniformly distributed in $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$. Test whether $X(t)$ is stationary or not? CO5 (5)

- b. If the Auto correlation function of a stationary process of $49 + \frac{6}{1+3\tau^2}$. Compute mean and variance. CO5 (6)

- B11.** If $R(\tau) = e^{-2.2|\tau|}$ is the autocorrelation function of a random process $X(t)$, compute the spectral density of $X(t)$. CO6 (10)

(OR)

- B12.** Compute the power spectral density of the random process $X(t) = A \cos(\omega t + \theta)$ where θ is uniformly distributed over $(0, 2\pi)$. CO6 (10)
